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ATHARNA

ATHARNA: Smart Museum Security & Visitor Experience System

Samsung Innovation Campus - Graduation Project

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Abstract

This project introduces ATHARNA, a smart IoT-based system designed to proactively protect museum artifacts and significantly enhance visitor experiences. It integrates multiple physical and environmental sensors (Load Cell, Ultrasonic, Gas, and DHT sensors), multi-layered authentication for staff and visitors, and a modern mobile application. The entire system is unified via the MQTT protocol (HiveMQ broker), enabling real-time data visualization on a Node-RED dashboard and a Flutter mobile app. This integrated approach provides continuous environmental monitoring, robust access control, and visitor management, marking an innovative step toward improved artifact safety and operational efficiency in modern museums.

1. Introduction

1.1. The Problem

Museums represent invaluable cultural and historical institutions that necessitate advanced security and environmental monitoring solutions. Traditional surveillance and manual monitoring methods are frequently insufficient to prevent theft, unauthorized handling of artifacts, or subtle environmental degradation.

Current security practices in many museums rely on outdated human supervision and simple alarms, which are prone to human error and often result in delayed response times in critical situations (such as theft or fire)⁷. Moreover, there is often a lack of integration between environmental monitoring systems and physical access control mechanisms, which can lead to inefficient operations and increased risk of loss or damage.

1.2. The Solution & Objectives

The Smart Museum Security & Visitor Experience System (ATHARNA) aims to combine modern IoT technology, computer vision, and mobile applications to offer a comprehensive and intelligent protection solution that addresses the shortcomings of conventional security setups.

The primary objectives of the ATHARNA project are:

- **Enhance Artifact Security:** Implement multi-sensor protection (weight and proximity) to detect unauthorized handling or removal in real-time.
- **Secure Staff Access Control:** Utilize **Facial Recognition (OpenCV)** to ensure only verified personnel (e.g., restoration staff) can approach or handle artifacts.
- **Environmental Safety:** Continuously monitor and log critical environmental parameters, including temperature, humidity (DHT11), and potential gas/fire hazards (MQ2).
- **Real-Time Data Access:** Provide immediate alerts and comprehensive data visualization through both a web-based dashboard (Node-RED) .
- **Improve Visitor Experience:** Offer visitors a streamlined experience through mobile booking, **NFC-based ticket entry**, and interactive artifact information via QR code scanning.



2. System Architecture and Methodology

The proposed system operates on a decentralized architecture, utilizing the Raspberry Pi as the central edge processing unit. Data communication across the network is standardized using the lightweight MQTT protocol with the HiveMQ broker for reliability.

2.1. System Components

Component	Category	Purpose
Raspberry Pi	Edge Controller	Main processing unit for sensor reading, Python logic, and OpenCV execution.
Weight Sensor (Load Cell)	Security	Detects artifact displacement or removal; crucial anti-theft trigger.
4x Ultrasonic Sensors	Security	Monitors proximity; creates a virtual protective barrier around the exhibit.
Gas Sensor (MQ2)	Environmental	Detects harmful gases and smoke/fire hazards; triggers physical alarm.
DHT11 Sensor	Environmental	Measures and transmits ambient temperature and humidity levels.
Camera Module	Computer Vision	Captures images for real-time Face Detection to verify authorized staff.
NFC Module	Visitor Access	Provides NFC-based ticket entry for visitors with valid mobile reservations.
MQTT (HiveMQ)	Communication	Ensures reliable, low-latency, real-time data exchange.
Node-RED Dashboard	Application	Visualization of all sensor data, visitor counts, and security alerts.
Flutter App	Application	Mobile interface for booking, cancellations, and QR-based artifact details.

2.2. Hardware & Access Control Flow

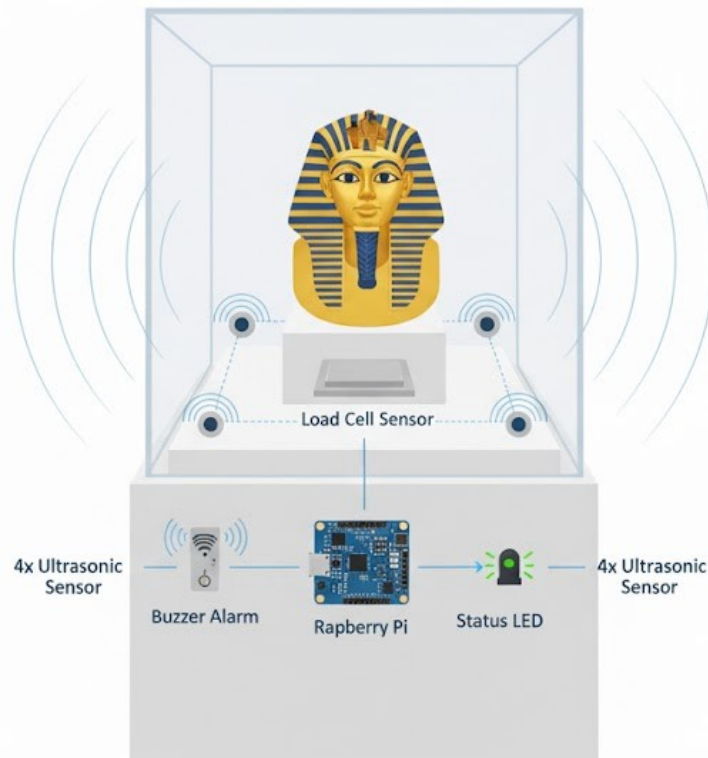
The hardware layer is designed for continuous, multi-parameter monitoring.

- **Artifact Integrity:** Each protected artifact is secured by both a Load Cell sensor (detecting lift) and four Ultrasonic sensors (detecting unauthorized close proximity).
- **Environmental Monitoring:** The DHT11 and MQ2 sensors are constantly sampling air quality. If the MQ2 detects a threat, a buzzer is immediately triggered.
- **Access Control Flow:** The **Camera Module** runs OpenCV to verify staff identity against an authorized database. The **NFC reader** grants entry/exit for visitors holding valid reservations. Unrecognized faces or unauthorized NFC tags trigger security



ATHARNA Smart Museum System

Physical Artifact Security Setup



protocols.

2.3. Software Layers

The software is structured into three distinct layers:

1. **Edge Layer (Raspberry Pi):** Python scripts handle sensor polling, signal processing, and the execution of the Face Recognition model. Cleaned data is then published to the Cloud Layer via MQTT.
2. **Cloud Layer (HiveMQ Broker + Firebase):** The MQTT broker facilitates all real-time communication. Firebase is used specifically to manage the Flutter app's data (user accounts, booking records).
3. **Application Layer (Node-RED + Flutter):** Node-RED subscribes to MQTT topics to display security and environmental data instantly. The Flutter app uses Firebase for transactional data (booking).

3. Application Layer & User Interface (UI/UX)

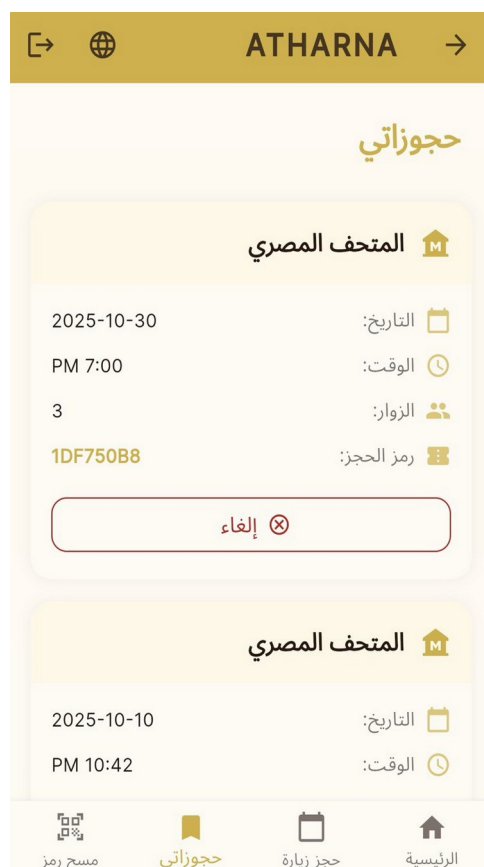
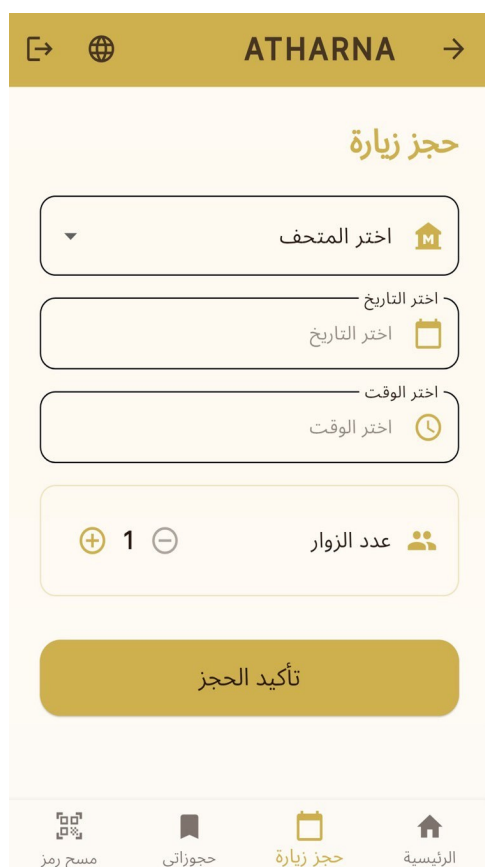
The Flutter mobile application provides a comprehensive and user-friendly interface for both visitors and administrators, designed with an intuitive flow to enhance the overall museum experience.

The image displays three screenshots of the ATHARNA mobile application interface, which features a clean, modern design with a gold and white color scheme.

Home Screen: The top bar shows the language selector (EN) and a globe icon. The main content area features the ATHARNA logo, the tagline "اكتشف حضارتنا بتقنية المستقبل" (Discover our civilization with future technology), and three buttons: "تسجيل الدخول" (Login), "إنشاء حساب" (Create Account), and "عن أثارنا" (About Us).

Login Screen: The top bar shows the title "تسجيل الدخول" (Login). The main content area features the ATHARNA logo, a form with fields for "البريد الإلكتروني" (Email) and "كلمة المرور" (Password), a "نسيت كلمة المرور؟" (Forgot password?) link, and a "تسجيل الدخول" (Login) button. At the bottom, there is a link "ليس لديك حساب؟ إنشاء حساب" (Don't have an account? Create account).

Registration Screen: The top bar shows the title "إنشاء حساب" (Create Account). The main content area features the ATHARNA logo, a form with fields for "الاسم" (Name), "البريد الإلكتروني" (Email), and "كلمة المرور" (Password), and an "إنشاء حساب" (Create Account) button. At the bottom, there is a link "لديك حساب بالفعل؟ تسجيل الدخول" (Already have an account? Login).



4. System Data Flow

1. Sensors collect data and transmit to the Raspberry Pi.
2. The Pi processes sensor data, executes Facial Recognition, and publishes all readings to the HiveMQ Broker via MQTT.
3. The HiveMQ Broker broadcasts the data to all subscribers: Node-RED.
4. Node-RED displays real-time readings, status, and logs security alerts.
5. The Flutter App handles user authentication and booking (via Firebase).

5. Results and Validation

The ATHARNA system achieved successful integration and functionality across all key modules:

- **Security Validation:** The combination of Load Cell and Ultrasonic sensors successfully detects both artifact removal and proximity violation.
- **Real-time Monitoring:** The system accurately reads environmental parameters, with smooth, real-time updates displayed on both the Node-RED dashboard.
- **Authentication:** The dual-factor security system effectively restricts staff access using **Face Recognition** and manages visitor entry via **NFC**.
- **Visitor Management:** The Flutter application provides a robust and user-friendly interface for booking, managing reservations, and accessing QR-based artifact information.

6. Future Enhancements

To further advance the ATHARNA system, the following enhancements are proposed:

- Integration with AI-based Threat Prediction Models for analyzing visitor movement patterns to detect suspicious behavior.
- Adding GPS-based tracking capabilities for mobile artifacts or staff assets.
- Expanding the database and optimizing the model for higher Facial Recognition accuracy and speed.
- Integrating with existing museum Enterprise Resource Planning (ERP) systems for full operational automation.

7. Conclusion

The Smart Museum Security & Visitor Experience System (ATHARNA) successfully meets its objectives by effectively combining IoT sensors, computer vision, and mobile technology to protect valuable museum artifacts⁵⁵. Through multi-layered authentication, continuous environmental monitoring, and intelligent real-time alerts, the system represents an innovative, practical, and highly scalable solution for creating smarter and safer museums.

8. References

- Internet of Things (IoT) Architecture for Smart Security Systems.
- Research Papers on Raspberry Pi-Based Surveillance and Access Control.
- MQTT Communication and HiveMQ Documentation.
- Node-RED and Flutter Official Documentation.